

Characterizing and Evaluating Working Emotion Vocabularies in Multilingual Large Language Models

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Abstract

Prior work evaluating emotion and affective understanding in large language models (LLMs) typically rely on predetermined label sets or focus on a singular evaluation task (e.g., emotion detection). We consider affective states, the much broader variety of terms people use to label their emotional experiences. We evaluate multilingual models’ understanding of affective states in English and Spanish through three different tasks: 1) identification where models predict an affective state given text, 2) expression, where models generate text expressing a given affective state, and 3) verification, where models report whether a given term refers to an affective state. We show that performance on one task is not necessarily predictive of performance on another. Using these three tasks, we then explore when and why models struggle to understand particular affective states compared to others. We identify systematic patterns in the affective state terms that are well and poorly understood by models, characterizing the working emotion vocabulary of LLMs.¹

1 Introduction

Benchmarks and approaches for emotion detection typically prescribe basic or universal psychological models of emotions (e.g., Ekman basic emotions (Ekman, 1984); emotion sets found in Wang et al. (2020). Recent work, however, has begun considering larger emotion sets (Demszky et al., 2020) as well as introduced benchmarks with greater coverage of world languages (Bianchi et al., 2022). Such studies are vital to ensuring that LLMs have a robust understanding of more nuanced and complex emotion expression in language.

Most recently, Deas et al. (2024) depart from traditional emotion detection and avoid a predetermined label set. Instead, they evaluate models’

¹We will make our code publicly available upon publication.

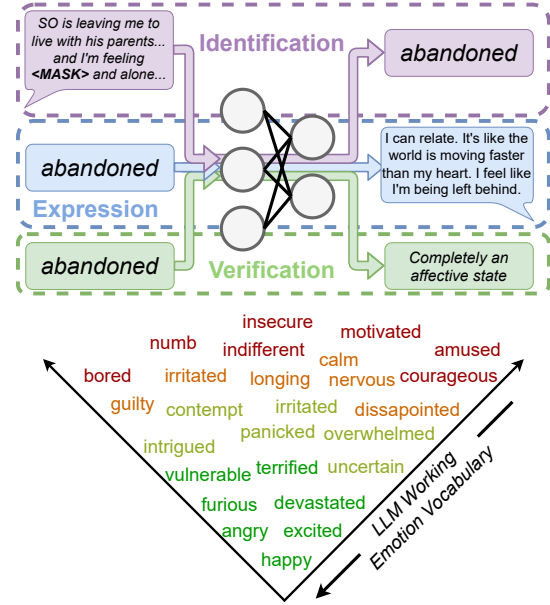


Figure 1: Through three tasks, affective state identification, expression, and verification, we seek to characterize the *working emotion vocabulary* of LLMs.

ability to predict any term used by an author to label their own feelings or emotions. The authors refer to these terms as *affective states*. In contrast to traditional emotion detection tasks, predicting affective states is shown to be difficult for recent large language models (LLMs) in both English and Spanish, warranting further investigation into *why* they exhibit poor performance in this setting. Furthermore, this and other prior work focus primarily on detecting emotion and affect in language, but we show that jointly considering other tasks and evaluation metrics reveals trends in model performance that were not otherwise apparent.

Accordingly, in this work, we investigate LLMs’ understanding of nuanced affective expression with multiple tasks to thoroughly analyze the affective states that models most and least understand. First, we evaluate models on *affective state identification* as in Deas et al. (2024). In contrast to identifying

affect in text, we additionally evaluate models on *affective state expression*, requiring models to generate text as if written by an author expressing a given affective state. Finally, we consider *affective state verification*, where models are expected to report the extent to which a given term in context refers to an affective state (e.g., compared to a physical sensation, "I feel sore").

Using these tasks, we analyze model performance on particular affective states. Specifically, we draw on psycholinguistics studies of *working emotion vocabularies*, the set of readily accessible terms available to someone in labeling their emotional experience (Schrauf and Sanchez, 2004). Beyond language and culture-level differences in the conceptualization of emotions, individuals can further differ in the salient subset of a language’s emotional lexicon that they tend to use at any given time. For example, one person may tend to use basic emotion labels (e.g., *happy*, *angry*, *sad*), while another may tend to use more precise terms (e.g., *ecstatic*, *enraged*, *discontent*) despite all terms being well understood by both individuals.

Similarly, despite being trained on text that likely includes a wide variety of emotional language, LLMs may show preferences or systematically superior understanding of particular affective states over others. These preferences may then hinder their understanding of affect in text written by diverse speakers. Therefore, we investigate LLMs’ understanding of nuanced affective expression in language and aim to characterize LLMs’ analogues to working emotion vocabularies. Due to the close relationship between language, cultural background, and emotion vocabularies, we specifically focus on multilingual LLMs and evaluate them across English and Spanish. We investigate the following research questions:

RQ1: *How well do multilingual LLMs’ understand nuanced affect in language across different tasks?* We benchmark 23 multilingual LLMs’ affective understanding by evaluating their ability to identify nuanced affect in text, express a given affective state, and verify whether a term refers to an affective state. We show that generally, **models struggle to predict nuanced affect (identification), instruction-tuned models particularly struggle to express given affective states (expression), and few models align with human judgments of what constitutes a typical affective state (verification).**

RQ2: *How does understanding of affective states vary across tasks and models?* We evalu-

ate whether understanding of different affective states is consistent across the three tasks for a given model, as well as whether there exist patterns across models. While we find consistent patterns across models, we also find that **there is little to no agreement among tasks, suggesting individual tasks provide a limited perspective on models’ affective understanding.**

Finally, **RQ3:** *What psycholinguistic factors characterize LLMs’ working emotion vocabulary?* We explore the extent to which different psycholinguistic properties (e.g., valence, arousal, age of acquisition) of affective states explain variation in LLM performance and behavior across tasks. We show that **age of acquisition**—the typical age a person learns a term—is **consistently significantly and positively predictive of model performance in English**, likely due to the composition of model pretraining corpora.

Overall, we show that relying solely on detection tasks provides an incomplete picture of models’ affective understanding, and that future work should consider a variety of diverse tasks to thoroughly characterize understanding. Furthermore, we use performance across our tasks to investigate the working emotion vocabulary of LLMs, identifying systematic patterns shedding light on when models succeed in and fail to understand affective states.

2 Background and Related Work.

Language and Emotions. Prior psycholinguistics work has distinguished terms that are considered *emotion-label words* (e.g., *happy*, *afraid*) which explicitly reference affective states, and *emotion-laden words* (e.g., *celebration*, *tragedy*) which express or elicit emotions without directly indicating a particular emotion (Pavlenko, 2008; Betancourt et al., 2024). Similarly, other work measures the prototypicality of emotion-label terms, considering how well a term represents the broader category comprising emotions (Pérez-Sánchez et al., 2021; Wu et al., 2025). Such distinctions, however—and accordingly, differences in individuals’ working emotion vocabulary—are realized differently according to cultural and linguistic background. For example, while distributions of positive, negative, and neutral emotion terms are largely similar across linguistic and cultural backgrounds (Bak and Altarriba, 2024; Schrauf and Sanchez, 2004), *frustration* has been found to be a more

salient *anger* term in English compared to other high-resource languages (Soriano and Ogarkova, 2025) and languages like Javanese can have notably diverse variation in terms for *sadness* compared to other languages (Suswandi and WS, 2017).

LLM Emotion Understanding. Traditionally, work in emotion analysis has focused on detection in language (e.g., (Demszky et al., 2020; Bianchi et al., 2022; Li et al., 2017) including in the evaluation of LLMs (Liu et al., 2024b; Deas et al., 2024; Boutouta et al., 2025). Rather than predicting emotions in text, some studies have investigated producing text that expresses an indicated emotion through style transfer and controllable generation (MohammadiBaghmolaei and Ahmadi, 2023; Xie and Agrawal, 2023), emotional dialogue generation (Liu et al., 2024a; Song et al., 2019; Ma and Chang, 2024), and other task contexts (Zhang et al., 2019; Ishikawa and Yoshino, 2025). Rather than restrict evaluation to a single task, one study has attempted to broadly evaluate emotional intelligence in LLMs (Sabour et al., 2024). These researchers’ study considers both emotion understanding (e.g., emotion detection) and emotional application (e.g., responses to emotion-laden dilemmas). In contrast to prior work, we focus on more comprehensive evaluation of models’ linguistic understanding of affective states. We leverage affective state terms derived from natural language and structure our evaluation to study individual terms.

3 Methods

3.1 Affective Understanding Tasks

We consider three different tasks capturing different aspects of affective understanding: affective state identification, verification, and expression. **Affective state identification** frames emotion detection as a generative task (Deas et al., 2024). Models are provided with input text where affective state terms are masked and tasked with predicting these terms used by the original author of the text. **Affective state verification** asks models to report whether an indicated term represents an affective state. Each term is provided with context to enable distinguishing cases that depend on the use of the term (e.g., "blue" as a color or as a descriptor of feeling). Finally, **affective state expression** requires models to generate a response to a given user prompt that expresses a target affective state. Illustrative examples of these tasks and their associated prompts are included in Figure 1 and Figure 2.

3.2 Data

For all tasks, we use the MASIVE dataset (Deas et al., 2024), consisting of English and Spanish Reddit posts that include explicit labels of the authors’ emotional experiences described (e.g., "I feel frustrated"). The researchers design an iterative procedure beginning with a seed set of affective state terms (e.g., the Ekman emotions (Ekman, 1984)) and a set of query templates (e.g., "I feel ____"). The templates used allow them to heuristically identify other affective state terms that appear in returned texts and include these terms in the next round of queries, resulting in a large set of unique affective states. The resulting data collected represents the discrete terms that people naturally use to label their own emotional experience. The text samples in this work were also annotated by 2 native English and Spanish speakers to identify whether the automatically detected terms represent affective states (e.g., rather than physical states, "I feel *sore*") in order to validate the data collection process.

Lang.	Split	Size	Len.	Score Dist. (%)			
				0	1	2	3
En	Exs	150	22.8	8.0	2.7	5.3	84.0
	Test	49	26.4	8.2	6.1	26.5	59.2
	Ovr	199	23.7	8.0	3.5	10.6	77.9
Es	Exs	150	63.8	27.3	3.3	8.0	61.3
	Test	50	68.5	20.0	2.0	22.0	56.0
	Ovr	200	65.0	25.5	3.0	11.5	60.0

Table 1: Summary data statistics with (rounded) percentage of words annotated within each score range by human annotators.

In particular, for evaluation, we leverage the subset of MASIVE with human judgments of whether a term reflects an affective state. Judgments are assigned on a 4-point Likert scale, where 0 and 3 represent "*not an affective state*" and "*completely an affective state*" respectively. We use the average judgments between annotators for overlapping samples. In experiments involving binary model outputs, we treat human annotator ratings ≥ 2 as positive (1) and < 2 as negative (0). Summary statistics and the distribution of labels is shown in Table 1 and examples of texts, automatically-identified candidate affective state terms, and the annotators’ ratings are shown in Table 2.

Expression Prompts. To evaluate models in the expression task, we form a dataset of input

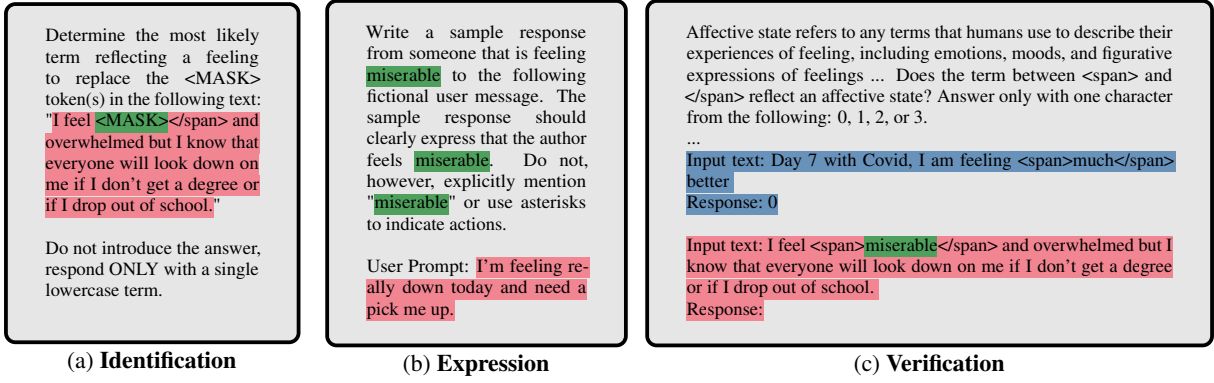


Figure 2: Condensed prompt templates for the three tasks. Red text represents input texts drawn from MASIVE, green text represents a masked or target affective state term, and blue text represents in-context examples in the verification task.

Lang	Context	Score
En	I feel jittery waking up but its probably left over anxiety.	1
	I'm so anxious, hoping I'm paranoid, but pissed off that I feel manipulated into being fearful vs. having a normal relationship with someone.	3
Es	Tengo una hipoteca. Odio a los corruptos, la falta de meritocracia y la injusticia. Estoy rodeado de gente en paro, de empresas que no han sido pagadas por los servicios prestados y de hipotecados.	0
	He tratado de hacerlo muchas veces pero nunca ha dado resultado, incluso llendo a bailar con pololas o amigos el resultado es el mismo, nose donde poner mis brazos ni como moverme, me siento incomodo	3

Table 2: Example texts and affect ratings on a 4pt-Likert scale, higher being "more like an affective state."

prompts to elicit responses where each is paired with an affective state that should be expressed in the response. For the input prompts, we use texts from the same subset of MASIVE as used in other experiments in order to encourage models to provide emotion-laden responses. We consider all possible pairs of input prompts and target affective states, allowing us to generate a diverse set of model responses for each affective state.

Working Emotion Vocabulary Measures. To characterize LLMs' working emotion vocabularies, we consider a variety of word-level characteristics and psycholinguistic norms collected by prior work. Namely, we consider the following measures that have been previously studied in the context of affect and language: valence and arousal (Mohammad, 2018; Stadthagen-Gonzalez et al., 2016), human age of acquisition (AoA) (Kuperman et al., 2012; Alonso et al., 2014), and concreteness² (Brysbaert et al., 2013). We additionally consider frequency in

²Concreteness is not included in the Spanish experiments due to the lack of coverage of the affective states we study in existing resources.

the July 2025 CommonCrawl snapshot measured using infinigram-mini (Xu et al., 2025) as a proxy of how often each term is seen during pretraining. We use these variables as inputs to regression models predicting model performance and behavior in each task in order to characterize working emotion vocabularies. Additional details on these measures can be found in Appendix A.

3.3 Experimental Configuration

Identification. We use the same setup and prompt as described in Deas et al. (2024), framing affective state identification as a masked span prediction task. As input, models are provided with a text where mentions of affective states by the author of that text are masked. Models are then tasked with filling each mask with the most likely term used by the original author to describe their feelings.

Expression. For affective state expression experiments, we instruct the model to generate a response to a provided user prompt and to ensure that the response clearly expresses a given affective state.³ An example of the prompt format is shown in Figure 2b.

Verification. Following prior work on the evaluation of values and opinions in LLMs (Röttger et al., 2024), we evaluate LLMs on the verification task with different evaluation formats. Specifically, we consider two output formats: binary (Yes/No) and ratings (Likert scores ranging from 0 to 3). An example of the rating format prompt is shown in Figure 2c. These formats reflect different levels of granularity in evaluating whether a given term is an affective state. We also explore two approaches to

³In earlier experiments, we found that passing target affective states into the system prompt is ignored by instruction-tuned models like OLMo 3. Therefore, we opt to include the instructions in a user prompt for all models.

processing predictions: label classification, where we select the relevant token with the highest likelihood, and a probability-based approach, where we compute a continuous score by averaging over possible output tokens (i.e., either yes/no or Likert ratings) weighed by their probability. This setup allows us to evaluate both discrete and probabilistic behavior. In summary, we test four variants: 1. **Binary (class)**: Most probable Yes/No response. 2. **Binary (prob)**: Weighted average over Yes/No responses. 3. **Rating (class)**: Most probable Likert-scale ratings. 4. **Rating (prob)**: Weighted average over Likert-scale ratings.

Finally, we evaluate models with few-shot prompting. We randomly sample human-annotated texts and ratings from MASIVE (*Exs.* in Table 1) as in-context examples, and include k -examples in each prompt for $k \in \{0, 1, 3, 5\}$. The few-shot examples for each input are held constant across models. Additional details on verification prompts are included in Appendix C.

3.4 Metrics.

Identification. We adopt the top- k similarity metric used in the original study (Deas et al., 2024) with $k \in \{1, 3, 5\}$. To account for affective state terms with context-dependent meanings (e.g., "blue"), we use a multilingual BERT model (Devlin et al., 2019) to produce contextualized embeddings, and extract the representation of each term within the context of the original text. Scores are then calculated as the maximum cosine similarity between the contextualized embeddings of the top- k predictions and ground truth terms.

Expression. To evaluate models in the expression task, we automatically detect the most likely affective state represented in each model-generated response and compare to the target affective state passed in the original prompt. To automatically detect affective states in each model response, we use mT5 models finetuned on the English and Spanish subsets of MASIVE (Deas et al., 2024) given that they were shown to outperform larger LLMs in identifying affective states. We extract the top-5 predictions from the corresponding mT5 model. We then compare the surfaced affective states from mT5 to the target affective state using the top- k similarity metrics described above. Overall, this results in a proxy measure of an LLMs' ability to express a given target affective state.

Verification. We compare model predictions against human annotations and compute agreement

using several metrics on each of the prompt variants. First, we compute Spearman's correlation on the output for each of the tasks and the scores assigned by the annotators. We additionally compute Cohen's κ for classification output formats. Throughout the presented results, we primarily focus on the correlation metric because it allows us to compare across task formats, while Cohen's κ is limited to these classification formats.

Working Emotion Vocabularies. We analyze the psycholinguistic characteristics that may explain variation in model understanding among affective states. In other words, we aim to identify classes of affective state terms that consistently result in high performance as well as those that result in low performance. To do so, we fit ordinal logistic regression models using affective state characteristics as predictors and a summary score of a model for each affective state across the three tasks as the response variable. To summarize model performance across tasks, we normalize each models' top-5 similarity scores in the identification task, weighted rating output in the verification task, and top-5 similarity scores in the expression task and take the average normalized score. Finally, we rank each affective state based on the aforementioned average score, with lower ranks indicating better performance. This allows us to identify patterns that may explain differences in models' capabilities to identify, verify, and express each affective state.

Through this regression model, the coefficients reveal the relationship between each psycholinguistic measure and model understanding: negative coefficients indicate an increase in a variable is associated with a decrease in affective state rank (i.e., an increase in model performance), while positive coefficients indicate that a decrease in a variable is associated with an increase in affective state rank (i.e., a decrease in model performance).

3.5 Models

We evaluate 23 multilingual models in total with sizes between 4 and 8 billion parameters (a full list is included in Appendix D). Specifically, we evaluate Llama, Bloom, Mistral, Deepseek, OLMO, and Qwen model families. To measure the impact of different training strategies, we consider both instruction and non-instruction-tuned models, as well as model variants finetuned for emotion analysis tasks (Liu et al., 2024b). We present results on a subset of 8 models throughout our main experiments for brevity, while results for remaining

models are presented in the Appendices.

4 Results

4.1 RQ1: Benchmarking Affective Understanding

	Model	Top- k Similarity $\uparrow$		
		1	3	5
English	OLMo 3 Base	37.1% $\pm$ 11.3	37.3% $\pm$ 11.4	37.3% $\pm$ 11.4
	OLMo 3	46.2% $\pm$ 19.4	58.6% $\pm$ 20.2	63.0% $\pm$ 19.9
	Llama 3.1 Base	30.3% $\pm$ 9.7	30.4% $\pm$ 9.6	30.4% $\pm$ 9.6
	Llama 3.1	44.8% $\pm$ 13.6	51.5% $\pm$ 15.2	56.0% $\pm$ 16.9
	Qwen 3 Base	34.2% $\pm$ 14.0	38.1% $\pm$ 17.2	38.3% $\pm$ 17.1
	Qwen 3	45.8% $\pm$ 13.9	55.8% $\pm$ 16.0	57.7% $\pm$ 16.4
	Ministral 3 Base	30.3% $\pm$ 8.4	30.8% $\pm$ 8.5	30.9% $\pm$ 8.5
	Ministral 3	43.3% $\pm$ 15.2	53.7% $\pm$ 16.9	58.4% $\pm$ 16.6
Spanish				
Spanish	OLMo 3 Base	30.7% $\pm$ 8.0	30.9% $\pm$ 8.1	30.9% $\pm$ 8.1
	OLMo 3	42.9% $\pm$ 14.2	49.4% $\pm$ 13.4	53.3% $\pm$ 14.4
	Llama 3.1 Base	39.8% $\pm$ 10.6	40.0% $\pm$ 10.7	40.1% $\pm$ 10.6
	Llama 3.1	44.3% $\pm$ 16.8	52.5% $\pm$ 17.2	56.8% $\pm$ 18.3
	Qwen 3 Base	37.7% $\pm$ 10.9	38.9% $\pm$ 11.0	40.2% $\pm$ 11.3
	Qwen 3	40.6% $\pm$ 15.5	49.5% $\pm$ 15.1	56.2% $\pm$ 17.4
	Ministral 3 Base	38.3% $\pm$ 10.8	38.4% $\pm$ 10.8	38.4% $\pm$ 10.8
	Ministral 3	45.7% $\pm$ 19.8	58.4% $\pm$ 20.3	60.9% $\pm$ 21.1

Table 3: Top- k similarity scores with $\pm$ 2SE’s for a subset of models in the identification task. Scores are shaded relative to those of other models for each value of k (darker indicating better performance).

Identification. We first evaluate model performance in affective state identification to gauge models’ capabilities in predicting affect expressed in text. Top- k similarity results for the identification task are shown in Table 3. We reaffirm the results in Deas et al. (2024) for a much larger set of LLMs: LLMs underperform the significantly smaller, finetuned mT5 model (71.1% and 78.1% top-5 similarity for English and Spanish respectively). Among the models, instruction-tuning appears to consistently improve performance for both English and Spanish. Common instruction datasets (e.g., Flan (Wei et al., 2022)) explicitly include sentiment and emotion tasks, which likely offers improvements in affective state identification performance. Full results are included in Appendix E.

Expression. As opposed to *only* identifying affective states given text, we additionally evaluate models’ ability to do the opposite – to generate text expressing a given affective state. Table 4 presents the top- k similarity results for each model in English and Spanish. Overall, the MASIVE-finetuned mT5 model is able to more accurately predict affective states in human-written texts compared to model-generated texts, suggesting models lag behind human ability to express affect in language. In contrast to other tasks, instruction-tuning does

	Model	Top- k Similarity $\uparrow$		
		1	3	5
English	OLMo 3 Base	42.5% $\pm$ 0.7	51.4% $\pm$ 0.7	55.6% $\pm$ 0.7
	OLMo 3	45.4% $\pm$ 0.6	55.3% $\pm$ 0.6	60.3% $\pm$ 0.6
	Llama 3.1 Base	42.7% $\pm$ 0.6	51.5% $\pm$ 0.6	55.5% $\pm$ 0.6
	Llama 3.1	38.4% $\pm$ 0.7	49.1% $\pm$ 0.7	54.1% $\pm$ 0.7
	Qwen 3 Base	40.1% $\pm$ 0.9	50.4% $\pm$ 0.9	54.3% $\pm$ 0.9
	Qwen 3	43.3% $\pm$ 0.6	53.1% $\pm$ 0.7	57.0% $\pm$ 0.7
	Ministral 3 Base	39.4% $\pm$ 0.9	50.3% $\pm$ 0.9	54.9% $\pm$ 0.9
	Ministral 3	42.1% $\pm$ 0.6	52.2% $\pm$ 0.7	56.1% $\pm$ 0.7
Spanish	OLMo 3 Base	38.2% $\pm$ 0.9	48.5% $\pm$ 0.9	53.5% $\pm$ 0.9
	OLMo 3	40.2% $\pm$ 0.8	50.8% $\pm$ 0.8	56.7% $\pm$ 0.9
	Llama 3.1 Base	38.9% $\pm$ 0.7	47.9% $\pm$ 0.7	52.8% $\pm$ 0.8
	Llama 3.1	36.3% $\pm$ 0.7	46.3% $\pm$ 0.8	51.4% $\pm$ 0.8
	Qwen 3 Base	45.3% $\pm$ 1.2	56.0% $\pm$ 1.1	61.2% $\pm$ 1.1
	Qwen 3	36.7% $\pm$ 0.8	48.7% $\pm$ 0.9	55.2% $\pm$ 0.9
	Ministral 3 Base	46.8% $\pm$ 1.2	57.6% $\pm$ 1.2	62.5% $\pm$ 1.1
	Ministral 3	34.2% $\pm$ 0.7	44.8% $\pm$ 0.8	50.4% $\pm$ 0.8

Table 4: Top- k similarity scores with $\pm$ 2SE’s for a subset of models in generating responses expressing a given affective state. Scores are shaded relative to those of other models for each value of k (darker indicating better performance).

not consistently lead to improved performance. Upon closer examination of model outputs, this may be due to instruction-tuning encouraging models to provide human-preferred responses to the given user prompt over expressing a given affective state. For example, the instruction-tuned models often provided empathetic or consoling responses to prompts mentioning negative emotions regardless of the affective state indicated. Full results are included in Appendix F.

Verification. To evaluate alignment between LLMs’ reports of whether terms represent affective states with human annotations, we consider the zero-shot performance across all models. Figure 3 summarizes the correlations across the four prompting formats. Few models (4-5 of those shown) exhibit significant correlations with human judgments with any prompting format in either language. Similarly to the identification task, models with agreement closest to the inter-annotator agreement baseline tend to be instruction-tuned models. Among prompting formats, relying on binary ratings leads to low or even negative correlations across models, suggesting coarse-grained reports may not be robust. Instead, model rating outputs—and particularly those that leverage output probabilities—appear more robust and more closely approximate human judgments for some models. Due to this, we use models’ rating probabilities for later analyses of behavior in the verification task. Full results including Cohen’s κ for classification evaluation formats are included in Appendix G.

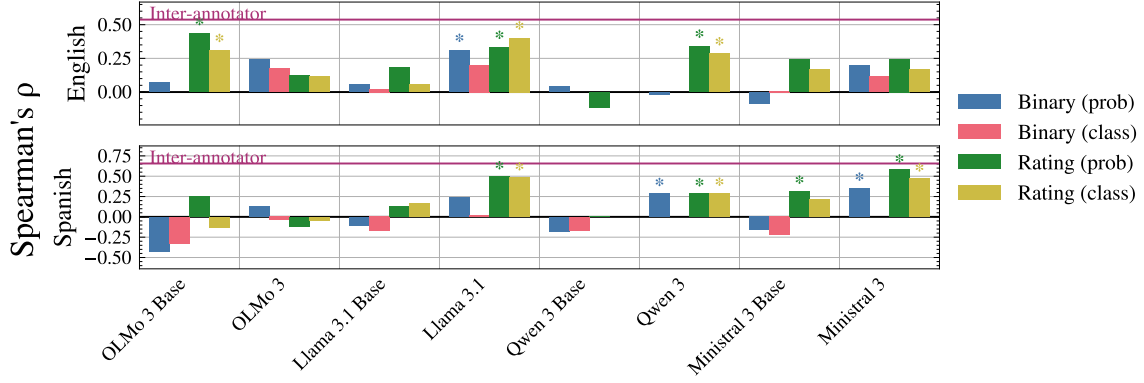


Figure 3: Spearman’s correlations between model outputs and human annotations across verification task formats. Agreement between annotators is indicated by the horizontal pink line. * indicates significant correlation ($p \leq .05$)

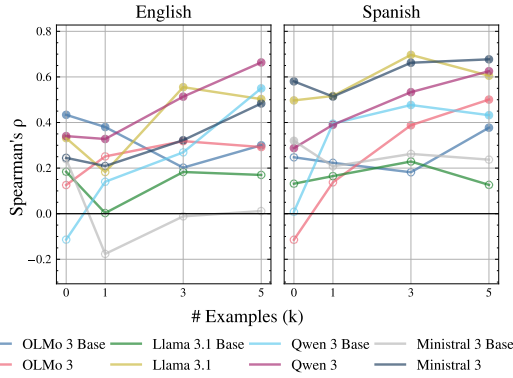


Figure 4: Spearman’s correlations between models’ predictions (rating probability) and human judgments in the verification task. Filled points $\bullet$ indicate significant correlations ($p \leq .05$) compared to unfilled $\circ$.

In Figure 4, we present changes in models’ performance across different numbers of in-context examples. For most models—namely, instruction-tuned models—few-shot examples tend to improve alignment with human judgments, in some cases, even exceeding inter-annotator agreement. This suggests that models have some ability to adapt to subjective human judgments in this task and can better approximate general notions of whether a given term represents an affective state. Given that our aim is to characterize the each models’ individual working emotion vocabularies, however, we focus on the 0-shot setting in later analyses.

4.2 RQ2: Model and Task Comparison

Table 6 presents the Spearman’s correlations at the affective state level between task performance measures for select models. While we might expect that performance is correlated across tasks for any given model, we find only weak to moderate correlations in select cases. Among pairs of tasks, identification and expression task performance ($I \times E$ in Table 6) appear consistently positively correlated ($\rho = .09$ to $.45$). The verification task, however, shows in-

consistent and rarely significant correlation with either the identification ($I \times V$, $\rho = -.06$ to $.29$) or expression ($V \times E$, $\rho = -.08$ to $.47$) tasks, supporting the lack of robustness of LLMs’ reports in this task. Furthermore, consistency across tasks varies among models and between languages, with Llama 3.1, for example, exhibiting positive correlations among all pairs of tasks in Spanish but little correlation in English. Overall, the lack of strong or consistent correlations suggests that individual tasks do not provide a complete evaluation of models’ affective understanding and that models should be evaluated from multiple perspectives.

We additionally qualitatively examine the affective states for which each model performs best and worst across tasks according to average rank (examples included in Appendix H.) Qualitatively, we see that models perform consistently well for select affective states (e.g., *abandoned* in English and *afortunada* in Spanish), even across languages (e.g., *nervous* and *nerviosa* appear highly ranked). In contrast, models perform consistently worse for terms that the original annotators judged to not represent affective states (e.g., *wide* and *jittery*). Given these qualitative trends observed across models, in the next section, we quantitatively analyze systematic patterns in affective state characteristics and model performance.

4.3 RQ3: Working Emotion Vocabularies

Regression coefficients for each psycholinguistic measure predicting model performance across tasks are shown in Table 5 for English and Spanish. The sign of coefficients is fairly consistent across models, suggesting these patterns may be largely independent of model architecture and training. Among predictors, for example, frequency (on average, $+.45$ for English) and arousal ($+.41$) positively pre-

English								
	OLMo 3 Base	OLMo 3	Llama 3.1 Base	Llama 3.1	Qwen 3 Base	Qwen 3	Ministral 3 Base	Ministral 3
Freq	0.62	0.02	0.91	0.21	1.11*	0.43	0.49	0.47
Valence	-0.56	-0.14	-0.20	-0.06	-0.84	-0.51	-0.80	-0.44
Arousal	0.51	0.51	0.37	0.30	0.83*	0.42	0.21	0.10
Conc	-0.28	0.23	0.09	0.28	-0.25	-0.07	-0.22	0.26
AoA	-1.13*	-0.95*	-1.00	-1.20*	-0.77	-1.28*	-0.73	-1.06*

Spanish								
	OLMo 3 Base	OLMo 3	Llama 3.1 Base	Llama 3.1	Qwen 3 Base	Qwen 3	Ministral 3 Base	Ministral 3
Freq	1.20	-1.72	0.01	-2.76	-2.53	-0.52	-0.48	-0.04
Valence	0.39	1.56	-0.28	1.57	0.85	2.29	1.21	1.23
Arousal	2.60	3.02	1.06	4.44	2.88	5.73 [†]	4.80	2.74
AoA	3.67	0.28	1.43	-0.72	-0.20	0.58	2.59	0.91

Table 5: Coefficients of the Ordinal Regression Model predicting affective state rank in English (top) and Spanish (bottom). Superscripts indicate significant coefficients, ** $p \leq .01$, * $p \leq .05$, [†] $p \leq .075$.

Model	I×V	I×E	V×E
English			
OLMo 3	0.11	0.39*	-0.01
Llama 3.1	0.04	0.12	-0.07
Qwen 3	0.08	0.40**	0.14
Ministral	-0.06	0.45**	-0.08
Spanish			
OLMo 3	0.10	0.15	0.18
Llama 3.1	0.29 [†]	0.30*	0.47**
Qwen 3	0.06	0.23	0.13
Ministral	0.14	0.09	0.35*

Table 6: Affective state-level Spearman’s correlations between performance measures in the Identification, Verification, Expression. Superscripts indicate significant correlations, ** $p \leq .01$, * $p \leq .05$, [†] $p \leq .075$.

dict model affective state rank (i.e., predict lower performance across tasks), while valence (-.43) and age of acquisition (-1.02) negatively predict rank (i.e., predict better performance across tasks). The only consistently *significant* predictor, however, is age of acquisition (AoA in Table 5) in English, suggesting that models tend to perform better on English affective state labels that humans learn later in life. These results oppose trends observed in human studies, which show a processing advantage for words acquired earlier in life (Wu et al., 2023). This trend may be explained by differences in human and model language acquisition, given that models are not necessarily trained on the typical style or amount of language seen during early human language acquisition (e.g., child-directed speech; Hago et al. 2024).

A closer examination of results for each task

show that age of acquisition’s effect is largely driven by model performance in the identification and expression tasks, while valence tends to be significantly predictive of models’ likelihood of rating a term as an affective state. In contrast to age of acquisition, the latter result does align with human studies which find processing advantages for positive words over negative words (Wu et al., 2023).

5 Conclusion

In this work, we evaluate multilingual LLMs’ understanding of affective states through three tasks: identifying affective states, generating text expressing a given affective state, and verifying whether a given term represents an affective state. We find that while instruction-tuning results in increased identification performance and alignment with human judgments in the verification task, instruction-tuned models struggle to generate text expressing a given affective state. Additionally, we find that model performance on one task is not consistently predictive of performance on others, suggesting that evaluations focusing on a singular task are inadequate to measure models’ understanding of affective states. Through these tasks, we then characterize the working emotion vocabulary of LLMs by studying psycholinguistic measures predictive of model performance and behavior. We find that in contrast to humans, age of acquisition is significantly positively predictive of affective state understanding across models likely due to differences in how humans and models acquire emotion lexicons.

Limitations

We note limitations accompanying our findings. First, we aim to more comprehensively evaluate models' understanding of affective states in natural language, but the tasks we consider are not exhaustive. We select three diverse tasks to capture different aspects of model understanding, although additional tasks and evaluation approaches may yield further insights into models' affective understanding. Additionally, we primarily find age of acquisition as a significant predictor of model performance in English, while other coefficients are rarely significant. The lack of significant coefficients does not indicate that no relationship exists, but our findings on the remaining relationships are inconclusive. Our experiments are limited by the availability of annotated texts from the MASIVE dataset, and we hope future work will pursue larger scale studies of these phenomena.

Ethics Statement

Our results reveal difficulties in models' understanding of affective states in natural language. Given the importance of understanding affect and emotions in critical settings such as mental healthcare, models' misunderstandings may lead to negative impacts on individuals emotional health when deployed. We focus on evaluating and characterizing models understanding of nuanced affect in language in order to anticipate potential harms resulting from misunderstanding. Given that working emotion vocabularies are specific to individual speakers, however, increased understanding of nuanced affective states may also further enable mass surveillance of specific communities online.

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Measure	Min	Max	μ	σ
Freq	4.51	9.02	6.34	1.13
Valence	-1.00	0.84	-0.43	0.47
Arousal	-0.96	0.90	0.11	0.47
Conc	1.46	4.07	2.52	0.57
AoA	3.60	13.41	8.24	2.76

Table 7: Summary statistics of psycholinguistic measures for the English affective states evaluated.

A Working Emotion Vocabulary Measures

We consider 5 measures in total: term frequency, valence and arousal (Mohammad, 2018; Stadthagen-Gonzalez et al., 2016), concreteness (Brysbaert et al., 2013), and age of acquisition (Kuperman et al., 2012; Alonso et al., 2014).⁴ Term frequency is measured through the count of each term in the July, 2025 CommonCrawl snapshot⁵ training dataset, and is collected using infinigram-mini (Xu et al., 2025) through the API⁶. All other terms are drawn from lexicons collected in prior literature. In cases where exact matches do not exist, we search for all variants of each affective state term, including verb conjugations for past participles and noun forms (e.g., we search for *abandon* and *abandonment* to represent the affective state *abandoned*). In Spanish, we additionally consider both masculine and feminine forms of each term.

Summary statistics of the psycholinguistic measures associated with each affective state in our working emotion vocabulary experiments are shown in Table 7 and Table 8 for English and Spanish respectively. For Spanish data, valence and arousal are originally measured on a scale of 1 to 9. In these tables, we rescale the values to range from -1 to 1 for clarity.

B Working Emotion Vocabulary Details

We fit Ordinal Logistic Regression models using the statsmodels package and the bfgs solver. All input measures except term frequency are normalized before fitting each model. For term frequency, we use the $\log_{10}(\text{frequency})$. As response variables, we first rescale model scores (top-5 similarity

⁴Citations refer to sources for each set of measures in English and Spanish respectively.

⁵<https://data.commoncrawl.org/crawl-data/CC-MAIN-2025-30/index.html>

⁶<https://infini-gram-mini.readthedocs.io/en/latest/api.html>

Measure	Min	Max	μ	σ
Freq	4.21	8.73	6.19	1.24
Valence	-0.87	0.79	-0.22	0.52
Arousal	-0.85	0.70	0.19	0.33
AoA	3.22	9.98	6.60	1.65

Table 8: Summary statistics of psycholinguistic measures for the Spanish affective states evaluated.

for identification, top-5 similarity for expression, and probability-based rating for verification) on each task to be between 0 and 1. We then group the average scaled performance across tasks into 5 quantiles. The ordered quantiles represent the response variable used in working emotion vocabulary regressions.

C Prompt Details

Full Prompt templates for the binary and rating-based verification task formats are included in Figure 5 and Figure 6 respectively. For non-instruction-tuned models, the system prompt is not provided.

Affective State Verification Prompt:

System: You are an expert in human emotions and feelings.

User: Affective state refers to any terms that humans use to describe their experiences of feeling, including emotions, moods, and figurative expressions of feelings (e.g. 'blue' as an expression of sadness instead of the color). Tell me if the word between and is an affective state or not. Output only Y if it is an affective state or N if it is not.

Do not explain or preface your answer.

Input text: Day 7 with Covid, I am feeling much better
Response: 0

Input text: I feel miserable and overwhelmed but I know that everyone will look down on me if I don't get a degree or if I drop out of school.
Response:

Figure 5: Full binary prompt template provided to models for affective state verification experiments.

D Model Details

All evaluated models are listed in Table 9 including whether the model is instruction-tuned, and whether it has undergone EmoLLM training (Liu

Affective State Verification Prompt:

System: You are an expert in human emotions and feelings.

User: Affective state refers to any terms that humans use to describe their experiences of feeling, including emotions, moods, and figurative expressions of feelings (e.g. 'blue' as an expression of sadness instead of the color). Does the term between and reflect an affective state? Answer only with one character from the following: 0, 1, 2, or 3.
0 means Not an affective state: the term does not refer to an emotion, feeling, or internal state.
1 means Unlike an affective state: the term is referencing something that is not an emotion.
2 means Like an affective state: the term is likely referencing an emotion, feeling, or internal state.
3 means Completely an affective state: the term is definitely an emotion, feeling, or internal state.
Do not explain or preface your answer.

Input text: Day 7 with Covid, I am feeling much better
Response: 0

Input text: I feel miserable and overwhelmed but I know that everyone will look down on me if I don't get a degree or if I drop out of school.
Response:

Figure 6: Full rating prompt template provided to models for affective state verification experiments.

et al., 2024b). In contrast to instruction-tuning, we do not find consistent benefits in EmoLLM training across experiments.

For the identification task we use beam search to generate the top-5 predictions for all models. For the expression task, we generate responses for all models using a temperature of 0.7, top-p of 0.9, and a no_repeat_ngram_size of 4. For verification experiments, we examine the raw output probabilities produced by the model for each possible label (i.e., "Y"/"N" or an integer 1-4).

E Full Identification Results

Results for all 23 models in the identification task are shown in Table 10.

F Full Expression Results

Results for all 23 models in the expression task are shown in Table 11.

G Full Verification Results

Results for all 23 models in the verification task are shown in Figure 7. Cohen’s κ metrics for classification formats are also included in Table 12.

H Affective State Rankings

The top and bottom 5 ranked affective states for 4 example models are shown in Table 13. we take the average ranking of each affective state compared to others across the three tasks.

Model	Checkpoint	Size	Training	
			Instruct	Emotion
OLMo (Groeneveld et al., 2024)	OLMo-7B-Instruct-hf	7B	✓	
OLMo-2 (Walsh et al., 2025)	OLMo-2-1124-7B-Instruct	7B	✓	
OLMo-3 (Olmo, 2025)	allenai/Olmo-3-1025-7B	7B	✓	
	allenai/Olmo-3-7B-Instruct	7B		
Llama-2 (Touvron et al., 2023)	Llama-2-7b-chat-hf	7B	✓	✓
	Emollama-chat-7b	7B	✓	
Llama-3 (Grattafiori et al., 2024)	Meta-Llama-3-8B	8B	✓	
	Meta-Llama-3-8B-Instruct	8B		
Llama-3.1 (Grattafiori et al., 2024)	Meta-Llama-3.1-8B	8B	✓	
	Meta-Llama-3.1-8B-Instruct	8B		
BLOOM (Workshop et al., 2023)	bloomz-7b1-mt	7B	✓	✓
	lzw1008/Emobloom-7b	7B	✓	
Mistral (Jiang et al., 2023)	Mistral-7B-v0.3	7B	✓	
	Mistral-7B-Instruct-v0.3	7B		
Ministral ⁷	Ministral-8B-Instruct-2410	8B	✓	
Ministral 3 ⁸	Ministral-3-8B-Base-2512	8B	✓	
	Ministral-3-8B-instruct-2512	8B		
Qwen-2.5 (Qwen et al., 2025)	Qwen2.5-7B	7B	✓	
	Qwen2.5-7B-Instruct-1M	7B		
Qwen 3 (Yang et al., 2025)	Qwen3-4B	4B	✓	
	Qwen3-4B-instruct-2507	4B		
DeepSeek-R1 (Guo et al., 2025)	DeepSeek-R1-Distill-Llama-8B	8B	✓	
	DeepSeek-R1-Distill-Qwen-7B	7B	✓	

Table 9: Summary of models evaluated throughout experiments. Emotion finetuned models are based on models released by Liu et al. (2024b).

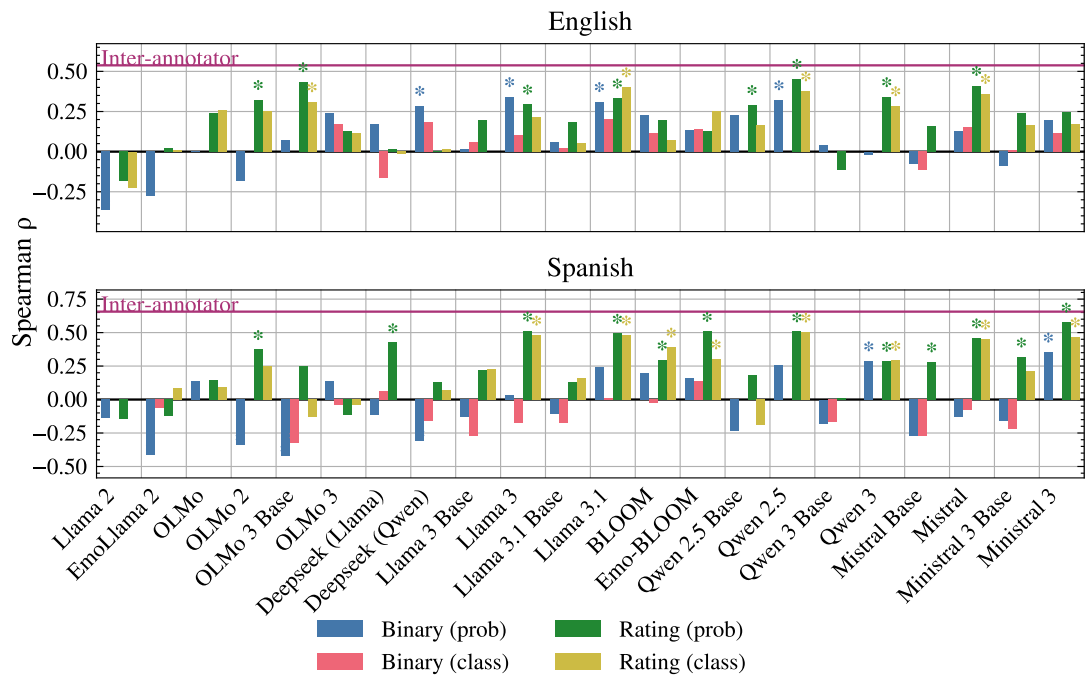


Figure 7: Full spearman's correlations between model outputs and human annotations across verification task formats. * indicates significant correlation ($p \leq .05$)

	Model	Top-k Similarity $\uparrow$		
		1	3	5
English	Llama 2	37.0% $\pm$ 13.8	40.4% $\pm$ 14.5	40.8% $\pm$ 14.6
	EmoLlama 2	39.2% $\pm$ 12.8	45.1% $\pm$ 16.5	45.4% $\pm$ 16.4
	OLMo	31.1% $\pm$ 10.7	33.9% $\pm$ 10.5	35.9% $\pm$ 11.8
	OLMo 2	44.2% $\pm$ 14.4	52.0% $\pm$ 14.7	56.4% $\pm$ 13.7
	OLMo 3 Base	37.1% $\pm$ 11.3	37.3% $\pm$ 11.4	37.3% $\pm$ 11.4
	OLMo 3	46.2% $\pm$ 19.4	58.6% $\pm$ 20.2	63.0% $\pm$ 19.9
	Deepseek (Llama)	25.2% $\pm$ 12.1	29.2% $\pm$ 11.2	30.2% $\pm$ 11.0
	Deepseek (Qwen)	24.5% $\pm$ 6.3	26.8% $\pm$ 6.9	28.7% $\pm$ 7.6
	Llama 3 Base	31.9% $\pm$ 9.2	32.1% $\pm$ 9.4	32.1% $\pm$ 9.4
	Llama 3	49.1% $\pm$ 17.5	56.7% $\pm$ 18.0	60.2% $\pm$ 18.1
	Llama 3.1 Base	30.3% $\pm$ 9.7	30.4% $\pm$ 9.6	30.4% $\pm$ 9.6
	Llama 3.1	44.8% $\pm$ 13.6	51.5% $\pm$ 15.2	56.0% $\pm$ 16.9
	BLOOM	36.9% $\pm$ 12.4	47.1% $\pm$ 13.7	51.5% $\pm$ 15.2
	Emo-BLOOM	40.8% $\pm$ 12.2	47.0% $\pm$ 12.6	50.2% $\pm$ 12.5
	Qwen 2.5 Base	27.0% $\pm$ 9.2	43.1% $\pm$ 14.1	46.0% $\pm$ 13.7
	Qwen 2.5	45.9% $\pm$ 15.7	57.8% $\pm$ 15.3	61.8% $\pm$ 16.2
	Qwen 3 Base	34.2% $\pm$ 14.0	38.1% $\pm$ 17.2	38.3% $\pm$ 17.1
	Qwen 3	45.8% $\pm$ 13.9	55.8% $\pm$ 16.0	57.7% $\pm$ 16.4
	Mistral Base	31.5% $\pm$ 11.6	32.5% $\pm$ 11.6	32.9% $\pm$ 11.9
	Mistral	42.0% $\pm$ 13.3	48.3% $\pm$ 13.7	52.0% $\pm$ 14.8
Spanish	Ministral	45.7% $\pm$ 15.9	53.1% $\pm$ 15.0	57.2% $\pm$ 14.4
	Ministral 3 Base	30.3% $\pm$ 8.4	30.8% $\pm$ 8.5	30.9% $\pm$ 8.5
	Ministral 3	43.3% $\pm$ 15.2	53.7% $\pm$ 16.9	58.4% $\pm$ 16.6
	Llama 2	39.6% $\pm$ 11.0	43.0% $\pm$ 11.2	43.7% $\pm$ 11.1
	EmoLlama 2	38.6% $\pm$ 10.2	47.1% $\pm$ 12.3	48.9% $\pm$ 13.4
	OLMo	36.9% $\pm$ 11.0	38.3% $\pm$ 11.0	38.9% $\pm$ 11.0
	OLMo 2	41.4% $\pm$ 12.9	52.5% $\pm$ 14.0	54.4% $\pm$ 14.7
	OLMo 3 Base	30.7% $\pm$ 8.0	30.9% $\pm$ 8.1	30.9% $\pm$ 8.1
	OLMo 3	42.9% $\pm$ 14.2	49.4% $\pm$ 13.4	53.3% $\pm$ 14.4
	Deepseek (Llama)	27.2% $\pm$ 8.8	30.6% $\pm$ 8.1	31.5% $\pm$ 7.7
	Deepseek (Qwen)	28.7% $\pm$ 8.2	31.6% $\pm$ 9.4	33.1% $\pm$ 9.4
	Llama 3 Base	38.2% $\pm$ 9.8	38.5% $\pm$ 10.0	38.7% $\pm$ 10.2
	Llama 3	46.5% $\pm$ 18.1	52.2% $\pm$ 17.1	56.7% $\pm$ 18.4
	Llama 3.1 Base	39.8% $\pm$ 10.6	40.0% $\pm$ 10.7	40.1% $\pm$ 10.6
	Llama 3.1	44.3% $\pm$ 16.8	52.5% $\pm$ 17.2	56.8% $\pm$ 18.3
	BLOOM	34.3% $\pm$ 12.1	40.1% $\pm$ 13.2	43.7% $\pm$ 16.0
	Emo-BLOOM	34.8% $\pm$ 12.1	41.4% $\pm$ 11.2	43.3% $\pm$ 11.2
	Qwen 2.5 Base	37.5% $\pm$ 13.3	44.2% $\pm$ 12.4	47.7% $\pm$ 14.8
	Qwen 2.5	40.8% $\pm$ 14.0	52.0% $\pm$ 16.8	61.1% $\pm$ 18.7
	Qwen 3 Base	37.7% $\pm$ 10.9	38.9% $\pm$ 11.0	40.2% $\pm$ 11.3
	Qwen 3	40.6% $\pm$ 15.5	49.5% $\pm$ 15.1	56.2% $\pm$ 17.4
	Mistral Base	31.0% $\pm$ 7.0	31.5% $\pm$ 7.0	31.5% $\pm$ 7.0
	Mistral	43.2% $\pm$ 12.9	49.0% $\pm$ 14.6	50.6% $\pm$ 14.5
	Ministral	42.2% $\pm$ 13.3	50.0% $\pm$ 14.6	52.3% $\pm$ 13.8
	Ministral 3 Base	38.3% $\pm$ 10.8	38.4% $\pm$ 10.8	38.4% $\pm$ 10.8
	Ministral 3	45.7% $\pm$ 19.8	58.4% $\pm$ 20.3	60.9% $\pm$ 21.1

Table 10: Full top-k similarity scores with $\pm$ 2SE’s for models in the identification task.

	Model	Top-k Similarity $\uparrow$		
		1	3	5
English	Llama 2	42.0% $\pm$ 0.8	52.6% $\pm$ 0.8	57.3% $\pm$ 0.8
	EmoLlama 2	40.4% $\pm$ 0.7	51.0% $\pm$ 0.8	55.5% $\pm$ 0.8
	OLMo	41.9% $\pm$ 0.7	51.0% $\pm$ 0.7	55.1% $\pm$ 0.7
	OLMo 2	41.8% $\pm$ 0.7	52.3% $\pm$ 0.7	56.7% $\pm$ 0.7
	OLMo 3 Base	42.5% $\pm$ 0.7	51.4% $\pm$ 0.7	55.6% $\pm$ 0.7
	OLMo 3	45.4% $\pm$ 0.6	55.3% $\pm$ 0.6	60.3% $\pm$ 0.6
	Deepseek (Llama)	38.2% $\pm$ 0.8	48.7% $\pm$ 0.8	52.7% $\pm$ 0.8
	Deepseek (Qwen)	34.6% $\pm$ 0.7	45.3% $\pm$ 0.8	49.2% $\pm$ 0.8
	Llama 3 Base	42.7% $\pm$ 0.6	51.7% $\pm$ 0.6	55.5% $\pm$ 0.6
	Llama 3	39.7% $\pm$ 0.7	49.7% $\pm$ 0.7	54.5% $\pm$ 0.7
	Llama 3.1 Base	42.7% $\pm$ 0.6	51.5% $\pm$ 0.6	55.5% $\pm$ 0.6
	Llama 3.1	38.4% $\pm$ 0.7	49.1% $\pm$ 0.7	54.1% $\pm$ 0.7
	BLOOM	41.9% $\pm$ 0.5	49.9% $\pm$ 0.5	53.4% $\pm$ 0.5
	Emo-BLOOM	41.4% $\pm$ 0.4	49.1% $\pm$ 0.5	52.6% $\pm$ 0.5
	Qwen 2.5 Base	44.0% $\pm$ 0.6	52.6% $\pm$ 0.6	56.3% $\pm$ 0.6
	Qwen 2.5	42.3% $\pm$ 0.6	51.4% $\pm$ 0.6	55.4% $\pm$ 0.6
	Qwen 3 Base	40.1% $\pm$ 0.9	50.4% $\pm$ 0.9	54.3% $\pm$ 0.9
	Qwen 3	43.3% $\pm$ 0.6	53.1% $\pm$ 0.7	57.0% $\pm$ 0.7
	Mistral Base	40.3% $\pm$ 0.8	49.9% $\pm$ 0.9	54.4% $\pm$ 0.9
	Mistral	40.6% $\pm$ 0.7	49.5% $\pm$ 0.6	52.9% $\pm$ 0.6
Spanish	Ministral	42.0% $\pm$ 0.7	52.1% $\pm$ 0.7	56.6% $\pm$ 0.7
	Ministral 3 Base	39.4% $\pm$ 0.9	50.3% $\pm$ 0.9	54.9% $\pm$ 0.9
	Ministral 3	42.1% $\pm$ 0.6	52.2% $\pm$ 0.7	56.1% $\pm$ 0.7
	Llama 2	36.0% $\pm$ 0.6	44.7% $\pm$ 0.7	49.6% $\pm$ 0.7
	EmoLlama 2	36.9% $\pm$ 0.6	45.8% $\pm$ 0.7	50.6% $\pm$ 0.8
	OLMo	38.1% $\pm$ 0.8	48.4% $\pm$ 0.9	53.8% $\pm$ 0.9
	OLMo 2	38.2% $\pm$ 0.7	47.5% $\pm$ 0.7	52.5% $\pm$ 0.8
	OLMo 3 Base	38.2% $\pm$ 0.9	48.5% $\pm$ 0.9	53.5% $\pm$ 0.9
	OLMo 3	40.2% $\pm$ 0.8	50.8% $\pm$ 0.8	56.7% $\pm$ 0.9
	Deepseek (Llama)	43.9% $\pm$ 1.1	54.9% $\pm$ 1.1	59.9% $\pm$ 1.1
	Deepseek (Qwen)	35.7% $\pm$ 0.8	45.3% $\pm$ 0.9	49.6% $\pm$ 0.9
	Llama 3 Base	39.6% $\pm$ 0.7	48.0% $\pm$ 0.7	52.4% $\pm$ 0.8
	Llama 3	35.8% $\pm$ 0.7	46.2% $\pm$ 0.8	51.3% $\pm$ 0.8
	Llama 3.1 Base	38.9% $\pm$ 0.7	47.9% $\pm$ 0.7	52.8% $\pm$ 0.8
	Llama 3.1	36.3% $\pm$ 0.7	46.3% $\pm$ 0.8	51.4% $\pm$ 0.8
	BLOOM	40.1% $\pm$ 0.6	47.0% $\pm$ 0.6	51.0% $\pm$ 0.6
	Emo-BLOOM	39.7% $\pm$ 0.5	46.8% $\pm$ 0.6	50.5% $\pm$ 0.6
	Qwen 2.5 Base	38.3% $\pm$ 0.7	47.2% $\pm$ 0.7	52.2% $\pm$ 0.8
	Qwen 2.5	38.2% $\pm$ 0.7	46.8% $\pm$ 0.7	52.1% $\pm$ 0.8
	Qwen 3 Base	45.3% $\pm$ 1.2	56.0% $\pm$ 1.1	61.2% $\pm$ 1.1
	Qwen 3	36.7% $\pm$ 0.8	48.7% $\pm$ 0.9	55.2% $\pm$ 0.9
	Mistral Base	41.6% $\pm$ 0.9	50.8% $\pm$ 1.0	55.3% $\pm$ 1.0
	Mistral	38.7% $\pm$ 0.7	48.9% $\pm$ 0.8	54.0% $\pm$ 0.8
	Ministral	40.1% $\pm$ 0.7	48.5% $\pm$ 0.7	53.5% $\pm$ 0.8
	Ministral 3 Base	46.8% $\pm$ 1.2	57.6% $\pm$ 1.2	62.5% $\pm$ 1.1
	Ministral 3	34.2% $\pm$ 0.7	44.8% $\pm$ 0.8	50.4% $\pm$ 0.8

Table 11: Full top-k similarity scores with $\pm$ 2SE’s for all models in generating responses expressing a given affective state.

Model	Format	English		Spanish	
		κ	κ_{bin}	κ	κ_{bin}
Llama 2	Binary	0.00		0.00	
	Rating	-0.05	-0.01	0.00	0.00
EmoLlama 2	Binary	0.00		0.01	
	Rating	-0.06	0.04	0.10	0.10
OLMo	Binary	0.00		0.00	
	Rating	0.07	0.00	-0.02	-0.07
OLMo 2	Binary	0.00		0.00	
	Rating	0.13	0.17	0.08	0.13
OLMo 3 Base	Binary	0.00		-0.13	
	Rating	0.22	-0.04	-0.07	0.00
OLMo 3	Binary	-0.12		-0.07	
	Rating	0.02	0.00	-0.03	0.00
Deepseek (Llama)	Binary	-0.07		0.01	
	Rating	-0.02	-0.08	0.00	0.00
Deepseek (Qwen)	Binary	-0.04		-0.16	
	Rating	0.05	0.06	0.03	0.10
Llama 3 Base	Binary	0.03		-0.08	
	Rating	0.00	0.00	0.03	0.13
Llama 3	Binary	-0.09		-0.07	
	Rating	0.25	0.00	0.22	0.00
Llama 3.1 Base	Binary	-0.06		-0.06	
	Rating	-0.04	-0.09	-0.02	0.06
Llama 3.1	Binary	0.04		-0.04	
	Rating	0.26	-0.09	0.22	0.42
BLOOM	Binary	0.01		-0.03	
	Rating	0.02	0.09	-0.03	0.38
Emo-BLOOM	Binary	0.01		0.00	
	Rating	-0.02	0.17	0.01	0.21
Qwen 2.5 Base	Binary	0.00		0.00	
	Rating	0.04	0.00	-0.06	0.00
Qwen 2.5	Binary	0.00		0.00	
	Rating	0.38	0.17	0.28	0.46
Qwen 3 Base	Binary	0.00		-0.04	
	Rating	0.00	0.00	0.00	0.00
Qwen 3	Binary	0.00		0.00	
	Rating	0.13	0.00	0.06	0.13
Mistral Base	Binary	-0.04		-0.12	
	Rating	0.00	0.00	0.00	0.00
Mistral	Binary	0.01		-0.10	
	Rating	0.23	0.00	0.18	0.00
Ministral	Binary	0.00		-0.08	
	Rating	0.00	0.00	0.00	0.00
Ministral 3 Base	Binary	0.01		-0.12	
	Rating	0.08	0.19	0.00	0.27
Ministral 3	Binary	0.01		0.00	
	Rating	0.19	0.00	0.22	0.47

Table 12: Cohen’s κ scores for classification output formats in the verification task. For κ_{bin} , ratings output by the model are binarized before computing agreement.

OLMo 3				Llama 3.1			
English		Spanish		English		Spanish	
Term	Rank	Term	Rank	Term	Rank	Term	Rank
abandoned	3.3	frustrada	5.7	lonely	5.0	afortunada	4.7
dizzy	9.0	afortunada	8.3	upset	6.3	frustrada	5.3
panicked	9.0	curioso	8.3	abandoned	7.7	nerviosa	9.0
lonely	9.3	triste	10.0	paralyzed	10.0	devastado	9.0
appalled	10.0	deprimida	11.0	unnoticed	13.7	deprimido	9.3
reluctant	30.7	rodeado	34.7	aloof	29.3	dormido	35.7
aloof	31.3	contra	37.0	done	29.7	rodeado	35.7
downtrodden	31.3	convencido	37.3	wide	29.7	fea	36.7
wide	33.3	obligado	38.3	jittery	31.0	trabado	38.0
jumpy	35.3	trabado	39.0	unbalanced	36.3	echo	39.3
Qwen 3				Ministral 3			
English		Spanish		English		Spanish	
Term	Rank	Term	Rank	Term	Rank	Term	Rank
abandoned	3.0	afortunada	6.7	abandoned	4.0	frustrada	7.7
upset	4.7	deprimida	9.7	lonely	5.7	afortunada	9.7
nervous	7.7	ignorada	10.3	supported	9.3	incomoda	11.3
aggressive	7.7	triste	10.3	deep	10.3	ignorada	12.3
lonely	7.7	frustrada	11.7	nervous	10.7	mejor	12.3
inexperienced	31.7	querido	34.3	aloof	30.3	contra	36.0
downtrodden	32.0	contra	34.7	downtrodden	30.7	mareado	37.7
reluctant	32.3	convencido	35.7	jittery	31.7	trabado	39.7
jittery	33.3	trabado	38.0	dizzy	31.7	convencido	40.0
aloof	35.3	rodeado	41.0	rested	34.3	rodeado	40.0

Table 13: Top and bottom-5 affective states by average performance rank across the three tasks. **Bolded** terms appear in the top/bottom ranks for all three models, while *italicized* terms appear for two.